

MATANZAS, Cuba

WMO: 782290

Lat: 23.034N

Lon: 81.435W

Elev: 64

StdP: 100.56

Time Zone: -5.00 (CUB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.4	13.2	7.8	6.6	14.6	9.1	7.2	16.3	9.4	22.3	8.5	22.3	1.7	160	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	33.1	24.8	32.2	24.8	32.0	24.8	27.0	30.8	26.5	30.4	26.1	30.1	3.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.5	29.8	25.2	20.5	29.1	25.0	20.2	29.0	85.2	31.0	83.2	30.6	81.4	30.3	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.7	6.9	DB	7.3	34.7	1.3	1.6	6.4	35.8	5.6	36.8	4.8	37.7	3.9	38.8	(4)
			WB	6.9	28.2	1.3	1.5	5.9	29.3	5.2	30.2	4.5	31.0	3.6	32.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.6	21.2	21.9	22.8	24.6	25.8	26.7	27.1	27.2	26.8	25.5	23.2	22.3
	DBStd	2.89	2.75	2.83	2.75	2.03	1.49	1.11	0.91	0.98	0.97	1.60	2.05	2.57
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	23	9	6	3	0	0	0	0	0	0	0	1	4
	CDD10.0	5331	348	334	397	438	488	502	529	534	503	481	396	382
	CDD18.3	2312	99	107	141	188	230	252	271	276	253	222	146	127
	CDH23.3	18444	488	684	1059	1501	1963	2226	2597	2579	2188	1659	831	667
	CDH26.7	6300	78	162	316	502	706	841	1059	1071	829	477	137	122
Wind	WSAvg	3.0	3.3	3.4	3.7	3.6	3.2	2.6	2.6	2.5	2.3	2.9	3.2	3.1
Precipitation	PrecAvg	1213	46	39	40	77	130	185	134	157	171	131	56	47
	PrecMax	1532	172	123	98	149	399	403	212	287	331	365	153	163
	PrecMin	918	4	7	18	24	39	81	62	95	77	37	2	7
	PrecStd	160	40	30	20	32	73	75	35	41	52	74	37	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	31.2	32.8	33.2	33.8	33.2	33.1	33.9	33.1	32.1	30.2	30.8
		MCWB	21.9	21.8	22.6	23.1	24.0	25.3	25.3	26.0	25.3	24.7	23.9	23.1
	2%	DB	28.6	30.0	31.2	32.0	32.1	32.2	32.2	32.9	32.1	31.0	29.2	29.1
		MCWB	21.7	21.8	22.2	22.9	23.8	25.3	25.4	25.3	25.3	24.9	23.4	22.9
	5%	DB	27.2	28.8	29.9	30.9	31.1	31.5	31.9	32.1	31.2	30.1	28.2	28.1
		MCWB	21.5	21.9	22.1	22.8	23.6	25.2	25.4	25.5	25.3	24.8	23.0	22.4
	10%	DB	26.1	27.2	28.7	29.8	30.2	30.9	31.1	31.2	30.8	29.2	27.2	26.9
		MCWB	21.0	21.4	21.8	22.6	23.4	25.2	25.4	25.5	25.3	24.3	22.3	22.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	24.4	25.1	26.6	26.8	27.5	27.5	27.7	27.0	26.6	25.6	25.1
		MCDB	27.5	27.9	29.8	30.2	30.4	31.1	31.1	31.6	30.6	29.8	28.7	27.9
	2%	WB	23.0	23.5	24.0	24.7	25.7	26.5	26.7	26.7	26.4	26.0	24.6	23.8
		MCDB	26.2	27.3	28.5	29.0	29.7	30.4	30.7	30.7	30.2	29.3	27.8	27.1
	5%	WB	22.3	22.7	23.2	23.7	24.9	26.0	26.2	26.3	26.0	25.5	23.9	23.2
		MCDB	25.7	26.5	27.8	28.6	29.4	29.9	30.3	30.4	29.9	28.9	26.9	26.5
	10%	WB	21.7	22.1	22.4	23.2	24.2	25.5	25.8	25.9	25.6	24.9	23.3	22.6
		MCDB	25.0	25.9	26.9	27.8	28.6	29.5	29.9	30.1	29.5	28.4	26.1	25.8
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.1	10.6	10.2	9.5	8.7	9.1	9.2	8.7	7.9	8.2	8.7
		MCDBR	10.2	11.1	11.9	11.4	10.6	9.5	9.7	9.9	9.3	8.6	8.7	9.6
		MCWBR	4.9	4.8	4.8	4.4	4.3	4.5	4.5	4.5	4.4	4.1	4.3	4.6
	5% WB	MCDBR	9.0	9.9	10.5	10.1	9.2	8.9	9.0	9.2	8.8	8.0	7.4	8.6
		MCWBR	4.8	4.8	5.0	4.7	4.5	4.7	4.8	4.7	4.6	4.3	4.2	4.6
Clear-Sky Solar Irradiance	taub	0.357	0.365	0.374	0.401	0.424	0.453	0.456	0.434	0.414	0.404	0.374	0.367	
	taud	2.538	2.517	2.478	2.400	2.376	2.345	2.328	2.410	2.461	2.473	2.507	2.528	
	Ebn at Noon	911	924	927	902	873	841	839	861	876	873	884	884	
	Edh at Noon	94	102	111	122	124	127	129	119	111	105	96	92	
All-Sky Solar Radiation	RadAvg	3.93	4.81	5.78	6.44	6.25	5.95	6.44	6.11	5.42	4.65	4.06	3.71	
	RadStd	0.30	0.29	0.22	0.20	0.62	0.42	0.32	0.23	0.28	0.32	0.18	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (11 neighbors)		+0.36	+0.96	N/A	N/A	N/A	N/A	N/A	-11	N/A	N/A

Nomenclature: See separate page